Introduction

In the last decades, cryptographic communities made many attempts in developing a digital currency able to be anonymously transacted between peers without the need of an intermediary (such as a bank).

Many attempts have been made such as “B-Money” by Wi Dai (1998), which was based on a Pub/Sub architecture requiring centralized servers, or “Bit Gold” theorized by Nick Szabo (1998) which was decentralized, and introduced a first version of a Proof-of-Work (PoW) consensus algorithm: computing power is spent to solve cryptographic puzzles, solutions are processed through a Byzantine Fault Tolerant (BFT) peer-to-peer network, and finally a cryptographic hash chain is created to link the most recent puzzle’s solution to the outcome of the following one, validating a block of transactions. This process of posting transactions to the network is consistent between Bitcoin and Bit Gold, although the latter did not make it to a usable implementation because of multiple security issues such as vulnerability to double-spending and Sybill attacks (a type of attack in which a party owning a huge number of nodes can perform a 51% attack independently of total hashrate).

In 2008 the release of the Bitcoin whitepaper by an anonymous collective known as “Satoshi Nakamoto” started the cryptocurrency revolution: it proposed a distributed ledger called Blockchain, on which each block is a group of transactions validated by each peer in a process called mining. In proof-of-work networks such as Bitcoin the first node to find a valid hash for the current block receives the block reward (a specific amount of newly issued coins which gets halved every fixed number of blocks) as well as fees paid on validated transactions in the block.

Since then many other cryptocurrencies have been developed, with different characteristics (For example, Ethereum aims to build a distributed virtual machine) and consensus algorithms such as Proof-of-Stake, in which the creator of the next block is chosen via various combinations of random selection and wealth or age (i.e., the stake) or Proof-of-Burn which virtualizes mining by a collateralization mechanism (ie “burn” Ethereum to mine DAI).

Marketplaces where these currencies can be exchanged between them or for real money called “Exchanges” began to flourish, transacted volume kept growing steadily and further exploded when Bitcoin reached its all-time high in December 2017.

Unlike traditional stocks trading, cryptocurrency exchanges were initially unregulated and far more accessible: paired with the wide spread on social media and news, the situation sparked a gold run. People looking for easy profit, professional traders as well as institutions all jumped on the “crypto train”.

Today, even with some initial regulations in place, capitalization is well above $400 billions, with more than $90 billion transacted each day and, albeit relatively young, we can affirm that crypto markets are inherently different than traditional stocks: prices are highly volatile and driven by demand and offer, with news and social media influencing (and amplifying) market moves, exchanges operate continuously without stops, and there are no rules regulating what market makers can or cannot do including pump and dump schemes and many others.

In this work we analysed historical data from various cryptocurrencies, based on supervised data mining techniques (classification) with the objective of generating trading signals spanning over multiple days to build a Decision Support System (DSS) for a human trader. Many studies on the subject have been carried out regarding traditional markets, while pertinence of these techniques on crypto markets is still a study subject.

Cryptocurrencies' nature would easily allow peer-to-peer SPOT trading, meaning a cryptocurrency can be exchanged with another one between peers who trust each other (otherwise a middleman called "escrow" is needed), although this is not the best way to earn profits from them. For this reason, Exchanges started to offer margin trading services, where orders are made against pools of collaterals, in which each user has a fixed allowance, usually depending on trading volumes. Margin trading allows two kinds of positions:

- LONG positions where the user borrows FIAT from the pool, buys collateral, then sells it when closing the position, paying back the initial debt and keeping the profit
- SHORT positions where the user borrows collateral from the pool, sells it for FIAT, then buys back the lent collateral when closing the position, keeping the profit.

Both types of margin orders expect the user to pay a fixed fee when position is opened, and a rolling fee depending on the position's length in time.

The proposed model is trained each day on N previous data points using a sliding window approach, aiming to find a strategy classifying data points in three classes, corresponding to generated signals:

- SELL if next day price variation is expected to be equal or below -1%
- HOLD if next day price variation is expected to be between -1% and +1%
- BUY if next day price variation is expected to be equal or above 1%

These signals are then fed into a test bed which simulates margin trading on a real exchange (with some relapsations which will be discussed later), opening positions with a 1% Stop Loss to minimize lost profit:

Signal	New Position	Existing Long	Existing Short
SELL	Open SHORT	Close LONG	Decrease Stop Loss
HOLD	Do Nothing	Do Nothing	If age > 1D then Close SHORT
BUY	Open LONG	Increase Stop Loss	Close SHORT

[Baselines]

[Results]

Source Data

Lack of regulation and different "age" for each cryptocurrency make it difficult to obtain a consistent amount of datapoints for all the examined cryptocurrencies. Another difficulty derives from the market's lack of regulation: exchanges usually reports time series using their own time zone, therefore data matching requires particular attention.

Yearly Volume and Candlestick data

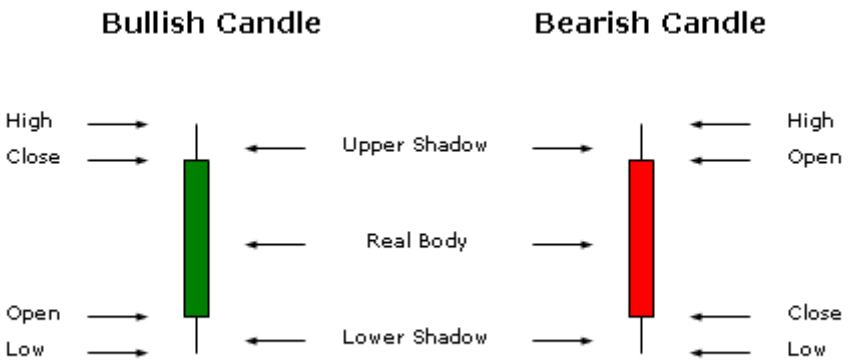
Candlestick data has been historically used since the 18th Century, when a Japanese man named Homma discovered that while there was a link between supply and demand in the price of rice, this was also influenced by traders' emotions and he did so by developing this kind of chart. This data representation derives from an aggregation of concluded orders from an orderbook (called Ticks) over defined intervals, assigning to each a tuple of values:

- Open: price at the beginning of said interval
- High: highest price in said interval
- Low: lowest price in said interval
- Close: price at the end of the interval



Figura 1 Bitcoin's Weekly Candlestick chart as of 02/12/2020

Candlestick data is visually represented as a sequence of bars, with each bar's colour depending on the difference between closing and opening price: if the price increased, the candle will be green (or empty) and called "Bullish", otherwise it will be red (or full) and called "Bearish". High and Low prices are represented respectively as "Wicks" and "Shadows", representing selling and buying pressure on the market.



Volume data is aggregated in the same way, and is the absolute sum of the units exchanged in the time interval.

This dataset was provided by PoliTO, and is composed of daily aggregated data in a pair of CSV files for each year, one including OHLC (Open, High, Low, Close) data for all the cryptocurrencies in that period, the other including Close and Volume (CV) data.

Blockchain data

Blockchain data would normally be freely available, by using specific tools such as blockchain-ETL (an acronym for Extract, Transform and Load) which decode its binary data structure and dump it to CSV format. Unfortunately, since Cryptocurrencies are developed by different groups of people with different principles they do not share the same data structures and implementation, therefore each would need a different ETL tool which is not always available. Also, since blockchains are usually very large and complex, these tools take huge amounts of time and processing power to run.

An option would have been using already extracted data provided by Google's BigQuery platform, but it did not cover most of the analysed assets.

Luckily, CoinMetrics' publicly available Community data covered most of the assets, offering a good range of key metrics for most of the analysed currencies, derived from both network and market activity while being consistent with our already acquired OHLCV data both in terms of timezone and granularity.

Such metrics are:

Feature name	Description
adractcnt	The sum count of unique addresses that were active in the network (either as a recipient or originator of a ledger change) that interval. All parties in a ledger change action (recipients and originators) are counted. Individual addresses are not double-counted if previously active.
blkcnt	The sum count of blocks created that interval that were included in the main (base) chain.
blksizebyte	The sum of the size (in bytes) of all blocks created that interval.
blksizemeanbyte	The mean size (in bytes) of all blocks created that day.
capmvrvcur	The ratio of the sum USD value of the current supply to the sum "realized" USD value of the current supply.
capmrktcurusd	The sum USD value of the current supply. Also referred to as network value or market capitalization.
caprealusd	The sum USD value based on the USD closing price on the day that a native unit last moved (i.e., last transacted) for all native units.
diffmean	The mean difficulty of finding a hash that meets the protocol-designated requirement (i.e., the difficulty of finding a new block) that interval. The requirement is unique to each applicable cryptocurrency protocol. Difficulty is adjusted periodically by the protocol as a function of how much hashing power is being deployed by miners.
feemeanntv	The mean fee per transaction in native units that interval.
feemeanusd	The USD value of the mean fee per transaction that interval.
feemedntv	The median fee per transaction in native units that interval.
feemedusd	The USD value of the median fee per transaction that interval.
feetotntv	The sum of all fees paid to miners that interval. Fees do not include new issuance.
feetotusd	The sum USD value of all fees paid to miners that interval. Fees do not include new issuance.
hashrate	The mean rate at which miners are solving hashes that interval. Hash rate is the speed at which computations are being completed across all miners in the network. The unit of measurement varies depending on the protocol.
isscontntv	The sum of new native units issued that interval. Only those native units that are issued by a protocol-mandated continuous emission schedule are included (i.e., units manually released from escrow or otherwise disbursed are not included).
isscontpctann	The percentage of new native units (continuous) issued over that interval, extrapolated to one year (i.e., multiplied by 365), and divided by the current supply at the end of that interval. Also referred to as the annual inflation rate.
isscontusd	The sum USD value of new native units issued that interval. Only those native units that are issued by a protocol-mandated continuous emission schedule are included (i.e., units manually released from escrow or otherwise disbursed are not included).
isstotntv	The sum of all new native units issued that interval.
isstotusd	The sum USD value of all new native units issued that interval.
nvtadj	The ratio of the network value (or market capitalization, current supply) divided by the adjusted transfer value. Also referred to as NVT.

nvtadj90	The ratio of the network value (or market capitalization, current supply) to the 90-day moving average of the adjusted transfer value. Also referred to as NVT.
pricebtc	The fixed closing price of the asset as of 00:00 UTC the following day (i.e., midnight UTC of the current day) for end-of-day data or the closest prior hour (nearest to that block) for block-by-block data, denominated in BTC.
priceusd	The fixed closing price of the asset as of 00:00 UTC the following day (i.e., midnight UTC of the current day) for end-of-day data or the closest prior hour (nearest to that block) for block-by-block data, denominated in USD.
roi1yr	The return on investment for the asset assuming a purchase 12 months prior.
roi30d	The return on investment for the asset assuming a purchase 30 days prior
splycur	The sum of all native units ever created and visible on the ledger (i.e., issued) as of that interval. For account-based protocols, only accounts with positive balances are counted.
txcnt	The sum count of transactions that interval. Transactions represent a bundle of intended actions to alter the ledger initiated by a user (human or machine). Transactions are counted whether they execute or not and whether they result in the transfer of native units or not (a transaction can result in no, one, or many transfers). Changes to the ledger mandated by the protocol (and not by a user) or post-launch new issuance issued by a founder or controlling entity are not included here.
txtfrcnt	The sum count of transfers that interval. Transfers represent movements of native units from one ledger entity to another distinct ledger entity. Only transfers that are the result of a transaction and that have a positive (non-zero) value are counted.
txtfrvaladjntv	The sum of native units transferred between distinct addresses that interval removing noise and certain artifacts.
txtfrvaladjusd	The USD value of the sum of native units transferred between distinct addresses that interval removing noise and certain artifacts.
txtfrvalmeanntv	The mean count of native units transferred per transaction (i.e., the mean "size" of a transaction) between distinct addresses that interval.
txtfrvalmeanusd	The sum USD value of native units transferred divided by the count of transfers (i.e., the mean "size" in USD of a transfer) between distinct addresses that interval.
txtfrvalmedntv	The median count of native units transferred per transfer (i.e., the median "size" of a transfer) between distinct addresses that interval.
txtfrvalmedusd	The median USD value transferred per transfer (i.e., the median "size" in USD of a transfer) between distinct addresses that interval.
txtfrvalntv	The sum of native units transferred (i.e., the aggregate "size" of all transfers) between distinct addresses that interval.
txtfrvalusd	The sum USD value of all native units transferred (i.e., the aggregate size in USD of all transfers) between distinct addresses that interval.
vtydayret180d	The 180D volatility, measured as the standard deviation of the natural log of daily returns over the past 180 days.
vtydayret30d	The 30D volatility, measured as the standard deviation of the natural log of daily returns over the past 30 days.
vtydayret60d	The 60D volatility, measured as the standard deviation of the natural log of daily returns over the past 60 days.

Social Media and Sentiment data

As said in the introduction, social media and user's sentiment play a crucial role in cryptocurrency markets, since their lack of regulation implies a direct cause-effect link between users' demand and market activity.

Unfortunately, most historical social media data available for cryptocurrencies is commercial, and the only freely available data of this kind is provided by CryptoCompare's api, but it only spans the last two years.

Included features are:

Feature name	Description
comments	The total number of posts/replies on the coin's CryptoCompare page.
posts	The number of posts on this coins CryptoCompare page.
followers	The number of users following this coin on CryptoCompare
points	Total points scored for this coin that day. Points are awarded for different metrics on each social media platform.
overview_page_views	Numbers of views for the 'Overview' tab on this coin's CryptoCompare page that day.
analysis_page_views	Number of views for the 'Analysis' tab on this coin's CryptoCompare page that day.
markets_page_views	Numbers of views for the 'Markets' tab on this coin's CryptoCompare page that day.
charts_page_views	Number of views for the 'Charts' tab on this coin's CryptoCompare page that day.
trades_page_views	Number of views for the 'Trades' tab on this coin's CryptoCompare page that day.
forum_page_views	Number of views for the 'Forum' tab on this coin's CryptoCompare page that day.
influence_page_views	Number of views for the 'Influence' tab on this coin's CryptoCompare page.
total_page_views	Total Number of page views for any tab on this coin's Cryptocompare page.
fb_likes	The number of users who had 'Liked' this page on Facebook as of that day.
fb_talking_about	The number of Facebook users 'Talking About' this coin (as determined by Facebook) that day.
twitter_followers	The number of followers for the requested coin's Twitter account as of that day.
twitter_lists	The number of Twitter lists the request coin featured in as of that day.
twitter_favorites	The number of times this coin's page had been favourited as of that day.
twitter_statuses	The number of tweets posted by the requested coin's Twitter account as of that day.
reddit_subscribers	The number of reddit users that subscribed to this coin's sub-reddit as of that day.
reddit_active_users	The number of 'active users' (as determined by Reddit) in this coin's sub-reddit that day.
reddit_posts_per_hour	The average number of posts per hour in this coin's sub-reddit as of that day.

reddit_posts_per_day	The average number of posts per day in this coin's sub-reddit as of that day.
reddit_comments_per_hour	The average number of comments per hour in this coin's sub-reddit as of that day.
reddit_comments_per_day	The average number of comments per day in this coin's sub-reddit as of that day.
code_repo_stars	The number of times this repository had been starred as of that day.
code_repo_forks	The number of forks splitting off from this repository as of that day.
code_repo_subscribers	The number of users subscribed to this repository as of that day.
code_repo_open_pull_requests	Number of issues with a pull request waiting on approval as of that day.
code_repo_closed_pull_issues	Number of issues with a pull request that have been closed as of that day.
code_repo_open_issues	Total number of open issues as of that day.
code_repo_closed_issues	Total number of issues closed as of that day.
code_repo_contributors	Total number of users who have contributed to this repo as of that day.

System Architecture

To perform the work, a testing bench has been written from scratch due to previous work being overly complex and tightly bound to stock markets.

The system is written in Python, leveraging the well-affirmed statsmodels, scipy and scikit-learn libraries as well as various extensions such as the XGBoost classifier.

Data Pre-processing

Raw data need to be pre-processed to be used in the system. Since each dataset has its own peculiarities, this process is different for each source and implemented in a custom script under the 'script/preprocessing' directory.

Common operations include dropping datapoints containing more than 5 missing values, initially filtering features by variance (Variance Thresholding) to remove those with null variance (because they would not add information to the model), and data manipulations such as splitting/merging both indices.

The results are stored in CSV/Excel files, containing all the datapoints for the relevant features per cryptocurrency, an index of the whole dataset (the set of said files) is stored in a json file in the form of {key:value} pairs, where key is the corresponding cryptocurrency symbol and value is a dictionary of the available file paths, indexed by type (csv or xls).

Dataset Building

Pre-processed datasets are further manipulated according to "processors": a processor is the "recipe" used to build a dataset, starting from the pre-processed data or another dataset. Processors implement the hypotheses behind the feature engineering step, and define the targets for classification (problem definition).

Many hypotheses have been implemented and tested in different scripts available under the 'processors' directory.

Datasets are stored in a pair of csv/excel files for each cryptocurrency, split between "features" and "target" and indexed in a json file like the pre-processed data. Additionally, a "feature_groups" key is added to the index, defining sub-groups of features which express a common characteristic (for example, volatility related technical indicators belong to the "volatility" group).

Problem definition

The first step in building a dataset is the definition of the problem the model will address; in other words this means defining the target feature or classification label.

Our target is to label each datapoint with different strategies, depending on the price variation expected on the next day. First, percent price variation is calculated by the formula:

$$close_pct(t) = (close(t) - close(t-1)) / close(t-1)$$

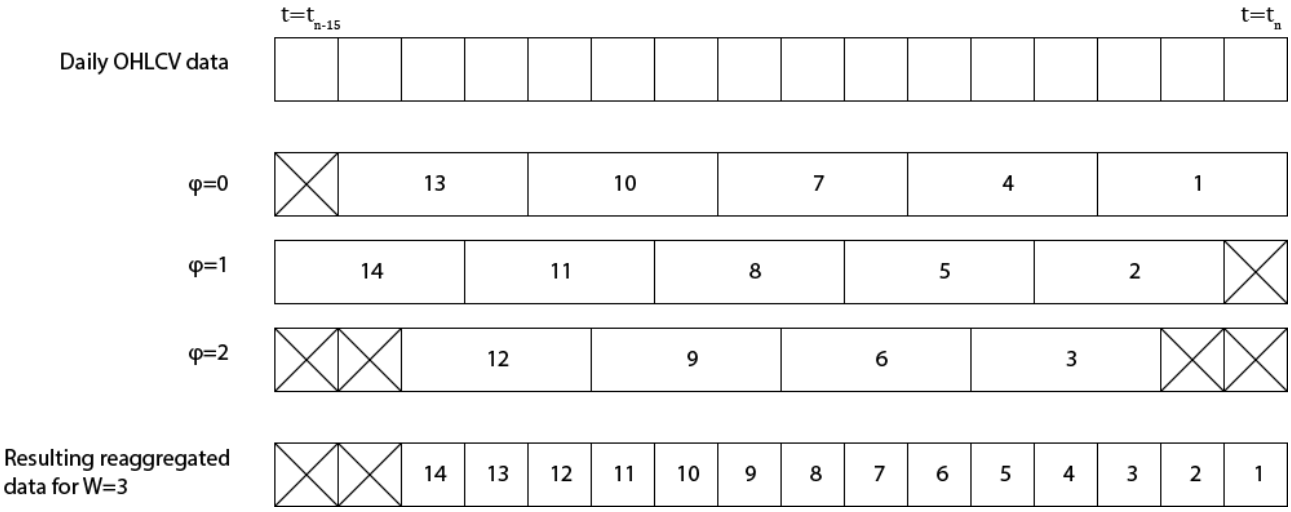
Then, the resulting time series is shifted backwards by one step, and the corresponding datapoint is labelled according to each of the following strategies:

Target Feature	# of Classes	Description
class	3	Obtained by thresholding next day price variation: if above or equal to 1% the datapoint is classified as BUY, if below or equal -1% the datapoint is classified as SELL, other cases are classified as HOLD

bin	3	Obtained by applying quantile separation to next day price variation: values falling below the 33% quantile are classified as SELL, cases above the 66% quantile as BUY, other cases are classified as HOLD
binary	2	Obtained by thresholding next day price variation in two classes: if above 0% is classified as BUY, if below 0% classification is SELL
binary_bin	2	Obtained by quantile separation to next day price variation: values falling below the 50% quantile are classified as SELL, other cases are classified as BUY

"Merged" Feature-set

For each symbol corresponding daily data is extracted, then resampled over W-periods wide intervals. The reaggregation is repeated W times, with a shift $\varphi = i$ added at each $i - th$ iteration in order to obtain a record for each period. At this point further metrics which depend on the sampling interval such as technical indicators are calculated, then the resulting data is merged to obtain the final features produced by this processor. The resampling process can be schematized as follows:



The resampling is applied to produce OHLCV and technical indicators data $for\ each\ w\ in\ [3, 7, 30]$, results are joined by date with original (daily) data and blockchain data. Technical indicators parameters are defined in json files, which are found under the 'config/ta_N.json' path where N is the aggregation interval's length in days.

"ATSA" Feature set

This processor builds the dataset described in the original work, obtained by joining OHLC daily data lagged with $W = 10$, and technical indicators:

Indicator	Description
rsma	Relative difference between a shorter and a longer moving average, according to the formula: $$RSMA(a,b) = \frac{\mathrm{SMA}(a) - \mathrm{SMA(b)}}{SMA(b)}$$
rema	Relative difference between a shorter and a longer exponential average, according to the formula: $$REMA(a,b) = \frac{\mathrm{RMA}(a) - \mathrm{EMA(b)}}{EMA(b)}$$

macd	Moving Average Convergence Divergence (MACD) is a trend-following momentum indicator that shows the relationship between two moving averages of a security's price. The MACD is calculated by subtracting the 26-period Exponential Moving Average (EMA) from the 12-period EMA.
ao	Aroon Oscillator is a trend-following indicator that uses aspects of the Aroon Indicator (Aroon Up and Aroon Down) to gauge the strength of a current trend and the likelihood that it will continue.
adx	The average directional index is an indicator used by some traders to determine the strength of a trend. The trend can be either up or down, and this is shown by two accompanying indicators, the Negative Directional Indicator (-DI) and the Positive Directional Indicator (+DI). ADX commonly includes three separate lines. These are used to help assess whether a trade should be taken long or short, or if a trade should be taken at all.
wd	Difference between Positive Directional Indicator (+DI) and Negative Directional Indicator (-DI) – see ADX. The price is moving up when +DI is above -DI, and the price is moving down when -DI is above +DI.
ppo	PPO is identical to the moving average convergence divergence (MACD) indicator, except the PPO measures percentage difference between two EMAs, while the MACD measures absolute (dollar) difference.
rsi	Relative Strength Index is a momentum indicator used in technical analysis that measures the magnitude of recent price changes to evaluate overbought or oversold conditions in the price of a stock or other asset. The indicator's values oscillate between 0 and 100.
mfi	The Money Flow Index (MFI) is a technical indicator that generates overbought or oversold signals using both prices and volume data. The oscillator moves between 0 and 100.
tsi	True Strength Index is a momentum oscillator which may be useful for determining overbought and oversold conditions, indicating potential trend direction changes via centerline or signal line crossovers, and warning of trend weakness through divergence.
stoch	Stoch is a momentum indicator comparing a particular closing price of a security to a range of its prices over a certain period with results in a 0-100 scale.
cmo	The Chande Momentum Oscillator is a momentum indicator like Wilder's Relative Strength Index (Wilder's RSI) and the Stochastic Oscillator. It measures momentum on both up and down days and does not smooth results, triggering more frequent oversold and overbought penetrations. The indicator oscillates between +100 and -100.
atrp	The average true range (ATR) is a technical analysis indicator that measures market volatility by decomposing the entire range of an asset price for that period. ATRP is this indicator's percent change w.r.t. previous period.
pvo	Percentage Volume Oscillator (PVO) is a momentum volume oscillator used in technical analysis to evaluate and measure volume surges and to compare trading volume to the average longer-term volume. PVO does not analyze price and it is based solely on volume. It compares fast and slow volume moving averages by showing how short-term volume differs from average volume over longer-term. Since it does not care a trend's factor in its calculation (only volume data are used) this technical indicator cannot be used alone to predict changes in a trend.
fi	The force index is a technical indicator that measures the amount of power used to move the price of an asset using price and volume. It is used for trend and

	breakout confirmation, as well as spotting potential turning points by looking for divergences.
adi	ADI Accumulation/distribution is a cumulative indicator that uses volume and price to assess whether a stock is being accumulated or distributed.
obv	On-balance volume (OBV) is a technical trading momentum indicator that uses volume flow to predict changes in stock price. Eventually, volume drives the price upward. At that point, larger investors begin to sell, and smaller investors begin buying. Despite being plotted on a price chart and measured numerically, the actual individual quantitative value of OBV is not relevant. The indicator itself is cumulative, while the time interval remains fixed by a dedicated starting point, meaning the real number value of OBV arbitrarily depends on the start date.

"Improved" Feature set

The idea for this set is to join lagged volume and close percent change data with $W = 10$, whose use is well-established, together with stats derived from OHLCV data and cherry-picked features engineered from Technical Analysis indicators and Blockchain data.

OHLCV derived features

Feature	Description
direction	Difference between closing and opening price. Sign represents the direction (upwards or downwards), while the modulo represents the magnitude.
day_range_pct	Percent variation of the difference between high and low price values w.r.t. the previous period. Should give insights about price volatility.

Technical Analysis features

Many features from the original ATSA work have been dropped because of high inter-correlation, as exposed by correlation matrices. Additionally, technical indicators are further processed to embed more significance in our context.

Feature	Description
rema_8_15_pct	Percent variation in the difference between 8-periods and 15-periods EMA, indicates changes in trend with better accuracy.
rsma_8_15_pct	Percent variation in the difference between 8-periods and 15-periods SMA, indicates changes in trend with better accuracy.
stoch_14_mean3_div100	Preliminary analysis proved this indicator to be too sensitive, having too many unique values. This has been addressed by evaluating a 3-period rolling mean, dividing the result by 100 to get a floating value.
macd_12_26_signal macd_12_26_pct	A nine-day EMA of the MACD called the "signal line," is usually plotted on top of the MACD line, which can function as a trigger for buy and sell signals. Traders may buy the security when the MACD crosses above its signal line and sell - or short - the security when the MACD crosses below the signal line. Moving Average Convergence Divergence (MACD) indicators can be interpreted in several ways, but the more common methods are crossovers, divergences, and rapid rises/falls. The first feature is the percent change of difference between MACD and its signal line, while the second is the percent change of the MACD (slope)
ppo_12_26_signal ppo_12_26_pct	PPO is interpreted in the same way as MACD, hence these features have the same significance as the MACD ones.
adi_pct adi_close_convergence	The accumulation/distribution measure seeks to identify divergences between the stock price and volume flow. This provides insight into how strong a trend is. If the price is rising but the indicator is falling this indicates

	that buying or accumulation volume may not be enough to support the price rise and a price decline could be forthcoming. The first feature reports the ADI percent variation (slope), while the second is a categorical feature indicating whether the indicator is converging or diverging with price change.
rsi_14_div100	RSI goes from 0 to 100, values <= 20 mean the asset is oversold indicating a good chance to BUY, while values >= 80 mean the asset is overbought, indicating a good chance to sell. It is divided by 100 to get a floating-point feature.
mfi_14_div100	An MFI reading above 80 is considered overbought and an MFI reading below 20 is considered oversold, although levels of 90 and 10 are also used as thresholds. A divergence between the indicator and price is noteworthy. For example, if the indicator is rising while the price is falling or flat, the price could start rising. It is divided by 100 to get a floating-point feature.
cmo_14_div100 cmo_14_signal	Many technical traders add a 10-period moving average to this oscillator to act as a signal line. The oscillator generates a bullish signal when it crosses above the moving average and a bearish signal when it drops below the moving average. The first feature represents the oscillator in a floating-point scale, while the second represents its relationship with the signal line.
obv_pct obv_mean3_pct	Usually, traders and analysts look to the nature of OBV movements over time. The slope of the OBV line carries all the weight of analysis, therefore the proposed features include percent change and percent change over a 3-periods rolling mean.
fi_13_pct fi_50_pct	Strong rallies in price should see the force index rise. During pullbacks and sideways movements, the force index will often fall because the volume and/or the size of the price moves gets smaller. Proposed features encode the slope of the indicator, calculated on 13 and 50 periods.
ao_14	AO moves between -100 and 100. A high oscillator value is an indication of an uptrend while a low oscillator value is an indication of a downtrend. Divided by 100 to get a floating-point scale.
atrp_14	Unchanged, it already is a percent value
pvo_12_26	Unchanged, it already is a percent value

Blockchain features

Indicator	Description
adractcnt_pct	Percent variation in number of active addresses, indicates changes in network activity w.r.t. previous interval.
earned_vs_transacted	Ratio representing miners' earning (given by the sum of issued coins and fees paid in that interval) divided by the sum of native units transferred between distinct addresses that interval.
isstot1_isstot365_pct	Percent variation in the ratio between issued coins and the rolling sum of the previous 365 days (starting from 7 days)
isstot1_splycur_pct	Percent change in the ratio between native unit issued in this interval and total supply
hashrate_pct	Percent change in network hashrate
roi30d	Untouched
isstotntv_pct	Percent change in issued native tokens
feetotntv_pct	Percent change in fees (miners reward)
txtfrcount_pct	Percent change in number of transactions

vtydayret30d	Untouched
isscontpctann	Untouched

"Timedspline" feature set

One of the problems arising from the "Improved" dataset was the high noise in prices' percent variation: this has been addressed by fitting an spline on such features, then considering the first and second derivatives, which represent the slope and convexity of the approximated line, instead.

Due to its nature spline fitting causes data leaks when done on the whole timeseries, care has been therefore taken to only fit the function on the datapoints preceding the current one. In other words, the feature value for day T is obtained by an spline which is fit on days $[0, T-1]$.

Indicator	Description
open_spl_d1 high_spl_d1 low_spl_d1 close_spl_d1	First derivative of the spline fit on the corresponding price from daily OHLCV data
open_spl_d2 high_spl_d2 low_spl_d2 close_spl_d2	Second derivative of the spline fit on the corresponding price from daily OHLCV data
candlestick_patterns_mean candlestick_patterns_sum	Mean and sum of scores obtained by applying all TALib's candlestick pattern recognition methods
close_volatility_7d close_volatility_30d	Price volatility. It is commonly deemed as a good indicator for cryptocurrencies' price because of psychological conditions influencing them such as FOMO (Fear of Missing Out) and sentiment in general. It is defined as the standard deviation of the rolling mean over a fixed period (respectively 7 and 30 days) of closing price's percent variation.
open_close_dist_pct	Percent variation of the candle's "body"
high_close_dist_pct	Percent variation of the candle's "wick"
low_close_dist_pct	Percent variation of the candle's "shadow"
high_low_dist_pct	Percent variation of day range (whole candle's height including wick and shadow)

Model Building and Parameter Search

Each of the used classifiers works in a different way requiring different input pre-processing (such as scaling and inputing). Therefore, classifiers have been separated from the building/testing logic using scikit-learn's "Pipeline" implementation and are located in the "pipelines" directory. Each pipeline exports:

- An "estimator" object, implementing sklearn's estimator interface, and is used for parameter search, evaluation, and trading
- A "PARAMETER_GRID" dictionary, containing the parameter grid used to perform Grid Search, specific for the classifier
- A "TARGET" attribute, indicating the preferred target for the pipeline (for example, if it proved better for that specific target)

Specific wrappers have been implemented and are provided in the library to transparently expose classifier's attirbutes to upper-level wrappers which require the model to have either "coef_" or "feature_importances_" attributes to perform feature selection such as SelectFromModel and RecursiveFeatureElimination.

Given a pipeline, each currency's dataset is split in a 70:30 train-test ratio, then grid search with 5-fold cross validation is performed to find optimal parameters (and features), optimizing the *precision* metric.

The "parameter search" step could be integrated in the sliding window approach to avoid introducing bias: we could use the data before the sliding window for this.

At the end of grid search, model is re-trained and performance is tested against the test set. A report is finally produced, including:

- Pickle dump of the model
- Plots of learning curves for the model (ROC_AUC and Precision/Recall) and Confusion Matrices
- Used training and test sets, along with feature names
- Found parameters
- Cross-Validation results (estimator, best_score, results)
- (Optionally, if exposed by the pipeline) feature_importances_ vector, support_ and ranking_

Model Benchmarking

Pipelines, datasets, and prediction targets are matched in a way like grid search to search for the optimal combination by parametrizing the existing model building script. Since the process takes a lot of time, progress and intermediate results are stored in a SQLite database so execution can be resumed in case of errors or power shortages. At the end of the process, a dataframe containing the results is exported: these can be useful because in the trading system each currency can use a strategy backed by a different classifier, picking the best one would likely result in better precision and therefore higher profit.

Trading Simulation

Unlike traditional assets, cryptocurrency markets do not have a closing time. Exchanges operate continuously, with the sole limitation of individual country regulation. This system can simulate both SPOT orders (which are executed between two or more users participating in an orderbook) and MARGIN orders (which are executed between the user and the Exchange's liquidity pools).

Exchanges usually impose an "allowance" to the liquidity that can be borrowed from the pools for margin orders, which depends on the user's "tier" who in turn depends on the volume traded on said exchange.

While SPOT order fees are independent of time and paid instantly in fiat over the opening/closing transaction, MARGIN orders incur both a fixed and a rolling (time-dependant) fee. The fixed fee is deducted from the lent amount, before purchasing the desired collateral, while rolling fees are paid when position is closed.

Trading simulation is done offline (paper trading), behind some assumptions:

- Ordered volume is negligible when compared to total traded volume, so
- Issued orders get instantly matched to an existing order in the orderbook, therefore Taker fees (which are usually higher) are paid
- Stop-loss is triggered immediately and assets in order are immediately sold at the stop loss price
- Margin pool allowances correspond to the average mid-tier on kraken
- Default fees as taken from kraken.com (one of the biggest European exchanges)
- Orders are placed and maintained at end of day (23:59:59) as soon as daily close is available

The trading system is composed of three parts:

- Strategy: trains a model (defined by pipeline) for a given day on given dataset with a sliding window approach, then predicting signals for the current day
- Exchange: handles fees and position operations (open, close), implementing the rules imposed by the exchange

- Backtest: Simulates day trading on a specific interval, trading on a given exchange instance using given strategies (one per each pair)

Strategy

This part takes care of training a given pipeline using given parameters, producing signals for a given cryptocurrency, on a given day.

The model is trained with a sliding window approach, with the window size depending on used classifier (for example, XGBoost classifier performs better with 150 training points).

To avoid training overhead for each backtest, trained pipeline instances are pickled (serialized in binary format) and saved with a specific pattern.

This unit produces "Signal" instances. A signal is a model with the following attributes:

Attribute	Type	Description
id	Integer (Auto-Increment)	An unique identifier for the model instance when saved in DB
day	DateTime	Day this signal refers to
symbol	Text	Currency symbol this signal refers to
signal	Integer (Enum)	Predicted label (SELL/HOLD/BUY)
label	Integer (Enum)	Expected label (SELL/HOLD/BUY)
close	Float	Closing price for the USD pair for this symbol

Exchange

Implements the exchange rules such as allowance for margin trades, asset management, funding, and fees.

This is the unit "doing the math", instantiating "Asset" and "Order" instances and filling their properties.

An order is a model with the following attributes:

Attribute	Type	Description
id	Integer (Auto-Increment)	An unique identifier for the model instance when saved in DB
symbol	Text	Symbol for the currency in order
type	Integer (Enum)	Order type (LONG/SHORT/SPOT)
status	Integer (Enum)	Order status (PENDING/OPEN/CLOSED)
coins	Float	Amount of coins (collateral) in order
open_price	Float	Price for the USD pair when order was opened
close_price	Float	Price for the USD pair when order was closed (NULL if open)
last_price	Float	Price for the USD pair when order was last checked
stop_loss	Float	Price for the USD pair when stop loss should be triggered
open_at	DateTime	Timestamp when order was opened
closed_at	DateTime	Timestamp when order was closed (NULL if open)
profit	Float	Profit for this order (NULL if open)
open_fee	Float	Fixed open fee in USD for this order (NULL if open)
close_fee	Float	Rolling fee in USD for this order (NULL if open)

An asset is a model with the following attributes:

Attribute	Type	Description
id	Integer (Auto-Increment)	An unique identifier for the model instance when saved in DB
symbol	Text	Currency symbol for the asset
coins	Float	Coins owned for this asset on SPOT wallet

fiat	Float	FIAT share for this asset on SPOT wallet
margin_fiat	Float	FIAT share for this asset on MARGIN wallet
margin_coins	Float	Coins share for this asset on MARGIN wallet
long_allowance	Float	Allowance for MARGIN LONG orders for this asset
short_allowance	Float	Allowange for MARGIN SHORT orders for this asset
long_orders	Integer	Number of LONG positions open on this asset
short_orders	Integer	Number of SHORT positions open on this asset
spot_orders	Integer	Number of SPOT positions open for this asset

Backtest

This unit is the core of the simulation, its role is to traverse each day in the period and, basing on the signals given by each strategy, act regarding position management.

Order sizing is implemented both in fixed fractional and fixed amount strategies:

- Fixed fractional attempts to invest a fixed percentage of the currency's equity value
- Fixed amount invests, as the name suggest, a fixed amount of fiat

Each day, for each currency a signal is generated using the corresponding strategy, then:

- Existing LONG positions are closed if the predicted signal is SELL, or stop loss is met (in that case, position is closed at stop loss price)
- Existing SHORT positions are closed If predicted signal is BUY, stop loss is met or position is older than 2 days and predicted signal is HOLD
- BUY signal causes the opening of a new MARGIN LONG position, with a -1% relative stop loss
- SELL signal implies opening a new MARGIN SHORT position with a +1% relative stop loss

At the end of each currency's operations, a new "Equity" model is instantiated and stored:

Attribute	Type	Description
id	Integer (Auto-Increment)	An unique identifier for the model instance when saved in DB
day	DateTime	Day this equity refers to
symbol	Text	Currency symbol
equity	Float	Sum of balances (both spot and margin) for the current asset, collateral balance is multiplied by last closing price